

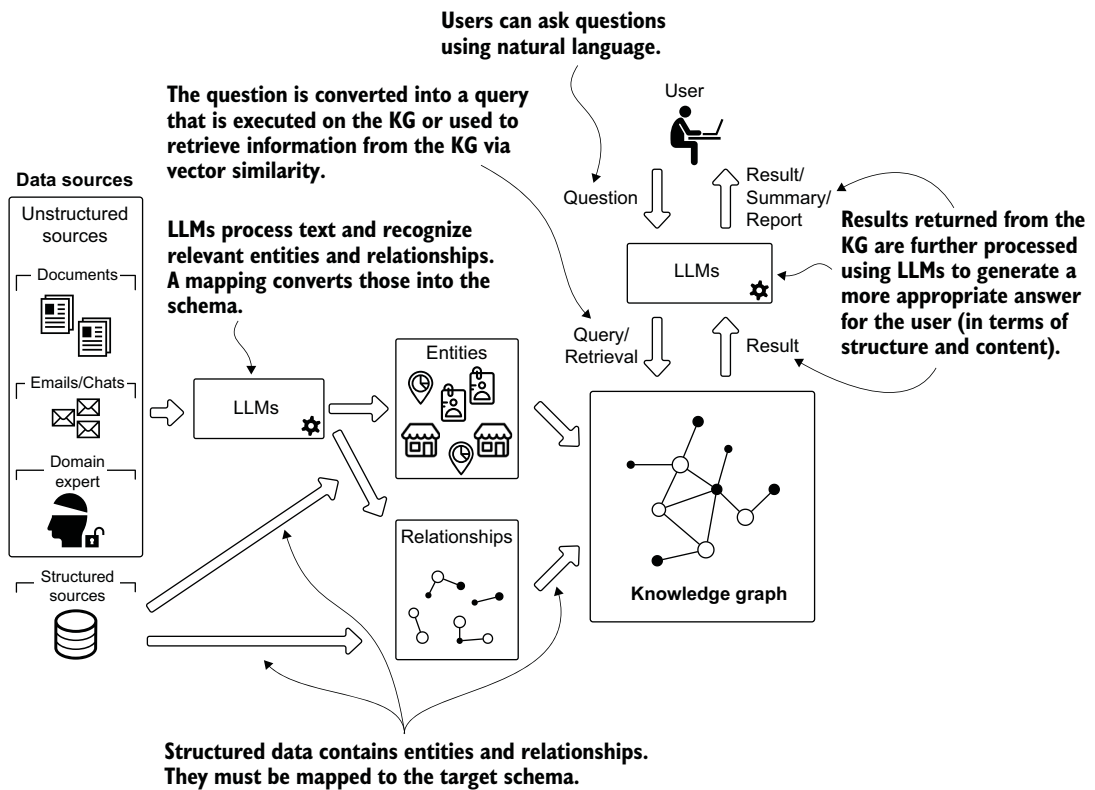
Knowledge Graphs and LLMs IN ACTION

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KGs Building from Structured and Unstructured Data, and LLMs Support for Querying and Retrieval



*Knowledge Graphs and LLMs
in Action*

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FOREWORDS BY MAXIME LABONNE
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To Aurora, Filippo, and Flavia
—Alessandro

*To my family—and especially my parents—for your unwavering love, support,
and patience. To my friends and mentors, for walking with me, inspiring me,
nudging me forward, and having the courage to be honest when it matters most.*
—Vlastimil

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timed reality checks have kept me grounded (and reasonably sane).*
—Giuseppe

*To my wife Fiorella and my children Giulio, Azzurra, and Arianna, who patiently
endured countless evenings of “I'm almost done writing.”*
—Fabio

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forewords

Working with graph neural networks and large language models over the years has taught me that each technology has profound strengths and equally profound limitations. Graph neural networks excel at understanding structured relationships but struggle with natural language interfaces. Large language models can engage in sophisticated conversations but frequently hallucinate facts and lack reliable grounding in structured knowledge.

Knowledge Graphs and LLMs in Action tackles an important challenge in AI: how do we combine these technologies to build systems that are both intelligent and trustworthy? Alessandro Negro, Giuseppe Futia, Vlastimil Kuš, and Fabio Montagna don't just theorize about this convergence; they provide practical recipes for making it work. Their approach bridges the gap between the precision of knowledge graphs and the accessibility of natural language, creating systems that can reason over complex data and explain their conclusions.

What impressed me most about this work is its rare emphasis on real-world implementation. The authors walk you through building knowledge graphs from messy, unstructured data and then show how to integrate them with language models for applications in healthcare, law enforcement, and beyond. The examples are concrete and the code is production-ready, making this both a learning resource and a practical guide.

The technical depth here is substantial, covering everything from graph construction to advanced retrieval systems, but the authors never lose sight of the practical goal: building AI systems that can serve as reliable advisors in critical decisions. This hybrid approach addresses the reliability and explainability challenges that have limited AI deployment in high-stakes environments.

If you're working on AI systems that need to be both powerful and trustworthy, *Knowledge Graphs and LLMs in Action* provides a clear framework for achieving it. The combination of knowledge graphs and language models represents a significant step toward AI that can handle complexity while maintaining the transparency and reliability that real-world applications demand.

—MAXIME LABONNE
HEAD OF POST-TRAINING, LIQUID AI

As a data science leader and passionate advocate for knowledge graphs, I'm thrilled to recommend *Knowledge Graphs and LLMs in Action*. We are witnessing a transformative moment in AI, shaped by the rise of generative AI and large language models (LLMs). Systems like Gemini and ChatGPT have opened the doors to natural language interaction at scale, offering a glimpse of intelligent machines. Yet we know these models are not without flaws. Hallucinations, outdated knowledge, limited transparency, and a lack of contextual grounding remain real challenges.

Addressing concerns like these is where knowledge graphs (KGs) shine, not just as a complement to LLMs, but as a necessary foundation for building accurate, explainable, and context-aware systems. This book demonstrates how the convergence of KGs and LLMs creates a powerful synergy, mitigating each other's weaknesses while unlocking their full potential.

The authors—Alessandro Negro, Vlastimil Kůs, Giuseppe Futia, and Fabio Montagna—bring years of hands-on experience and consulting expertise. Their work moves beyond theory to deliver actionable, production-ready insights grounded in real-world applications.

This book is more than a reference for knowledge graphs and LLMs. It's a practical toolkit for developing intelligent systems that enhance, not replace, human decision-making across domains like healthcare, finance, and law enforcement.

In an age where AI must be transparent, contextual, and trustworthy, this book is both timely and essential. It belongs on the shelf of every data scientist, engineer, architect, and knowledge-driven professional ready to build the next generation of intelligent systems.

Thank you, Alessandro, Vlastimil, Giuseppe, and Fabio, for this insightful and practical book!

—KHALIFEH ALJADDA
DIRECTOR OF DATA SCIENCE, GOOGLE CUSTOMER SOLUTIONS

preface

When I was nearing completion of my previous book, *Graph-Powered Machine Learning*, I reached out to my acquisitions editor, Mike Stephens, with a proposal for a natural continuation. That earlier work introduced knowledge graphs and demonstrated how they could be built using natural language processing, but many readers pointed out that graph neural networks were a significant missing piece. My proposed book would fill that gap while extending the knowledge graph story further, including detailed analysis and building techniques.

Mike accepted the proposal, and I embarked on a new adventure with the working title *Knowledge Graphs Applied*. Recognizing the scope of the challenge, I invited three colleagues from GraphAware—Fabio, Giuseppe, and Vlastimil—to join the effort, confident that their combined expertise would be invaluable. I naively thought that if one author could write a book in four years, four authors could complete a book in just a year. That assumption proved as flawed as expecting nine women to deliver a baby in one month.

Reality had other plans. Over the past years, significant changes swept through the technology landscape. Large language models (LLMs) and generative AI disrupted the field entirely, and knowledge graph practitioners suddenly found themselves with unprecedented opportunities to use this established technology in revolutionary ways. We initially planned to build on existing natural language processing (NLP) tools like BERT, but these were rapidly being superseded by LLM capabilities that opened new possibilities for building, querying, and analyzing knowledge graphs.

This was precisely where many practitioners, ourselves included, were struggling. Rather than resist this transformation, we decided, together with Mike and Dustin Archibald (our development editor), to embrace it. We adjusted our title to *Knowledge Graphs and LLMs in Action* and substantially revised the content to position LLMs as an

integral component of our ultimate goal: intelligent advisor systems that empower humans in performing complex decision-making tasks. This pivot required extensive refactoring and a fundamental shift in our approach, but the result exceeded our expectations.

The book you are reading has evolved into a manifesto for the power of hybrid systems. It demonstrates how combining these technologies—knowledge graphs, which are well established, and LLMs, which are newly emerged—creates a flywheel effect that delivers remarkable long-term results. Knowledge graph practitioners will discover how to use LLM capabilities for greater impact, and LLM practitioners will learn techniques that address some of the major limitations of language models.

We invite you to join us on this journey toward more intelligent, more reliable, and more human-centered AI systems.

acknowledgments

This book took almost five years to complete, and during that time, many things changed around us, both professionally and personally. The technology landscape has transformed dramatically since we began writing: LLMs have fundamentally shaped our professions.

Writing a book requires dedication and countless hours, usually outside normal working hours, late at night, and on weekends and holidays. So first and foremost, we need to thank our families and all the people who somehow received a “no” or suffered a delay because of this book.

To my co-authors—Fabio, Giuseppe, and Vlastimil—thank you for embarking on this adventure with me. Each of you brought unique expertise and perspectives that made this book infinitely better than what I could have accomplished alone. Your dedication to excellence and willingness to adapt as the technology landscape shifted around us were nothing short of remarkable.

We owe an enormous debt of gratitude to the team at Manning Publications. In particular, Mike Stephens, Manning’s associate publisher, not only accepted our book proposal but also provided invaluable guidance during the pivotal transformation from our original direction (*Knowledge Graphs Applied*) to what you are now reading (*Knowledge Graphs and LLMs in Action*). A special thank you to Dustin Archibald, our development editor, who followed us step by step with enormous patience and consistently provided excellent advice to make this a better book; your commitment to quality and your understanding during the extensive refactoring process made all the difference. We also want to thank the production and marketing teams at Manning—there are so many talented individuals that it’s impossible to mention them all, but they are the reason Manning books are such high quality and so well presented.

Our sincere appreciation goes to Dimitris Polychronopoulos, our technical editor, whose meticulous attention to detail and expert feedback significantly improved the technical accuracy and clarity of this work. Dimitris is an R&D scientist and entrepreneur specializing in genomics and data-driven drug discovery. With roles across biotech and big pharma, he has led innovative work on applying knowledge graphs and AI to uncover novel targets in oncology and chronic liver diseases.

We also extend our gratitude to all the reviewers who provided valuable feedback throughout the development process, including those who shared comments and suggestions online: Alexey Ott, Angelo Simone Scotto, Ayush Tomar, Avinash Tiwari, Chalamayya Batchu, Charles Ivie, Chris Viner, Dan McCreary, David Cronkite, David Meza, Floris Bouchot, Gajendra Babu Thokala, Gourav Sengupta, Guillaume Alleon, James J Byleckie, Jeremy Chen, Kristof Leroux, Kumar Abhishek, Lawrence Nderu, Maria Ana, Nicolas Bievre, Or Golan, Ozan Evkaya, Pethuru Raj, R. P. Shrivastava, Richard Vaughan, Robert Wardenga, S. S. Narendran, Sachin Panemangalore, Samantha Berk, Shailja Gupta, Simeon Leyzerzon, Sophia Shvets, Sumit Pal, and Suvarsha Rai. Your insights and constructive criticism helped shape this book into its final form.

A special acknowledgment goes to Khalifeh and Maxime, who kindly agreed to write forewords for this book. They received an almost-final copy when the book was ready for production, requiring them to work under tight deadlines to provide the thoughtful forewords you'll find at the beginning—no small feat for professionals as busy as they are. Khalifeh and Maxime are among the most knowledgeable experts we know in their fields. Their endorsement carries particular weight because of their extensive experience in bringing these technologies to real-world applications, and their ongoing work continues to inspire us and the broader community.

We also want to acknowledge the companies that provided us with the knowledge, experience, and opportunities we needed to create this book. GraphAware, in particular, has been instrumental in shaping our understanding of real-world graph applications and the challenges organizations face when implementing these technologies at scale.

Finally, we extend our appreciation to the broader community of researchers, practitioners, and open source contributors whose work made many of our examples possible. The datasets, tools, and frameworks that power the demonstrations in this book represent countless hours of effort from dedicated individuals who chose to share their knowledge with the world.

This book exists because of all of you. Thank you for making this journey possible.

about this book

Knowledge Graphs and LLMs in Action is a comprehensive guide to building hybrid intelligent systems that combine the structured reasoning capabilities of knowledge graphs (KGs) with the natural language understanding of large language models (LLMs). This book demonstrates how these complementary technologies can work together to create more powerful, reliable, and explainable AI solutions that address real-world challenges across various domains.

Who should read this book

This book is designed for machine learning engineers, data scientists, graph experts, and AI engineers who want to harness the synergistic power of KGs and LLMs. Whether you're working with structured enterprise data, building recommendation systems, developing fraud detection algorithms, or creating question-answering applications, this book will show you how to use both technologies to achieve better results than either could deliver alone.

If you're a data scientist looking to enhance your models with structured knowledge, a machine learning engineer seeking to reduce hallucinations in LLM applications, or an AI practitioner interested in building explainable and verifiable systems, this book provides the practical guidance you need. Although some familiarity with machine learning concepts and graph databases is helpful, the book introduces all necessary concepts and builds complexity gradually.

How this book is organized: A roadmap

The book has 15 chapters organized into 5 parts, progressing from foundational concepts to advanced implementations.

Part 1 establishes the theoretical and practical foundations for hybrid intelligent systems:

- Chapter 1 introduces the powerful combination of KGs and LLMs, demonstrating their complementary nature through concrete examples and use cases.
- Chapter 2 explores fundamental concepts of intelligent systems, diving deep into knowledge representation and reasoning strategies, and illustrating how KGs and LLMs work together in practice.

Part 2 focuses on building KGs from structured data sources:

- Chapter 3 demonstrates KG construction through a healthcare example, showing how to help clinicians diagnose rare diseases using the Human Phenotype Ontology.
- Chapter 4 expands on these foundations with advanced analysis methodologies, including community detection algorithms and multisource integration across biomedical applications.

Part 3 tackles the challenging realm of extracting knowledge from unstructured text:

- Chapter 5 demonstrates the fundamental pipeline for converting text to KGs using both traditional natural language processing (NLP) and modern LLM-based methods through a case study at the Rockefeller Archive Center.
- Chapter 6 expands on document processing workflows, from OCR scanning to sophisticated graph analytics for identifying research networks and influence patterns.
- Chapter 7 explores named entity disambiguation in healthcare regulation, showing how to link entities to structured knowledge bases like the Unified Medical Language System (UMLS) and Systematized Nomenclature of Medicine (SNOMED).
- Chapter 8 introduces an innovative approach to disambiguation that combines open LLMs with domain ontologies for enhanced accuracy.

Part 4 explores machine learning applications on KGs:

- Chapter 9 introduces fundamental concepts for applying machine learning to KGs and establishes the theoretical foundation for learnable representations.
- Chapter 10 illustrates feature engineering approaches through practical examples in fraud detection and drug repurposing.
- Chapter 11 advances into graph neural networks, showing how these architectures automatically learn optimal representations from graph structures.
- Chapter 12 demonstrates real-world applications through anti-money laundering and movie recommendation systems.

Part 5 brings everything together in practical information retrieval systems:

- Chapter 13 explores integrating KGs with LLMs through retrieval augmented generation, demonstrating graph RAG systems.

- Chapter 14 shows how to build sophisticated question-answering systems that emulate domain expert reasoning through a law enforcement example.
- Chapter 15 provides a complete implementation using LangGraph and Streamlit, demonstrating how to build production-ready systems.

The book is designed to be read sequentially for comprehensive understanding, but experienced practitioners can focus on specific parts based on their immediate needs. Beginners should start with parts 1 and 2 to establish foundational knowledge before exploring specialized applications in later parts.

About the code

This book contains extensive source code examples demonstrating practical implementations of KG and LLM integration. Code appears both in numbered listings and in line with explanatory text, formatted in a fixed-width font like this to distinguish it from regular content. Sometimes code is also in **bold** to highlight relevant pieces.

In many cases, the original source code has been reformatted; we've added line breaks and reworked indentation to accommodate the available page space in the book. In rare cases, even this was not enough, and listings include line-continuation markers (↪). Additionally, comments in the source code have often been removed from the listings when the code is described in the text. Code annotations accompany many of the listings, highlighting important concepts.

All source code examples are available for download from the book's GitHub repository at <https://github.com/alenegro81/knowledge-graphs-and-llms-in-action>. The repository includes complete implementations for each chapter's examples, along with setup instructions and data files needed to run the code.

The examples require Python 3.8 or higher and various libraries, including Neo4j, NetworkX, transformers, LangChain, and Streamlit. Specific requirements and installation instructions are provided in each chapter and the repository's documentation. Some examples also require access to OpenAI APIs or other language model services, with instructions for setting up the necessary credentials.

You can get executable snippets of code from the liveBook (online) version of this book at <https://livebook.manning.com/book/knowledge-graphs-and-llms-in-action>. The complete code for the examples in the book is also available for download from the Manning website at <https://www.manning.com/books/knowledge-graphs-and-llms-in-action>.

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about the authors



ALESSANDRO NEGRO is the chief scientist at GraphAware, where he supervises the science and technology area responsible for delivering Hume, a mission-critical analytics platform that uses knowledge graphs (KGs) at its core. He holds a Ph.D. in computer science and has successfully deployed machine learning systems combined with graphs for numerous organizations across various industries. Dr.

Negro is the author of *Graph-Powered Machine Learning* (Manning, 2021). His recent work focuses on integrating LLMs with KGs to create more reliable and explainable AI systems at scale. Beyond his role at GraphAware, Alessandro actively mentors and advises startups, helping organizations in specialized domains create custom models tailored to their unique requirements.



GIUSEPPE FUTIA is a data scientist and research fellow with expertise in knowledge graphs, large language models, and graph neural networks. He holds a Ph.D. in computer engineering and bridges academia and industry through extensive experience in research and applied innovation. His work focuses on developing knowledge-driven intelligent systems that integrate symbolic and statistical AI to

enable advanced decision-making across diverse domains.



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FABIO MONTAGNA is a senior graph and AI engineer at GraphAware, with more than 15 years of experience in research-driven software engineering across neuroscience, operational oceanography, and natural language processing. As a pioneer in bridging scientific discovery and commercial application, he specializes in transforming complex research concepts into scalable production systems that deliver business value. His dual role as product strategist and technical expert enables him to translate theoretical breakthroughs into practical solutions across healthcare, earth sciences, and commercial applications. Beyond GraphAware, Fabio has been collaborating with neurologists in pharmaceutical research for Parkinson's disease treatment evaluation.

about the cover illustration

The image featured on the cover of this book is called “Mařatanka” by Jořa Uprka (1861–1940), a Czech painter renowned for his vivid portrayals of folk life in southern Moravia, a historic wine-growing region in the Czech Republic. The painting depicts a young woman from the village of Mařatice, whose name echoes the ancient designation “sons of Maria” for its inhabitants. Dressed in a traditional local costume, she carries a tray of grapes and fruit aloft, embodying the spirit of harvest, abundance, and the enduring rhythm of historical rural life. It is a celebration of shared European traditions rooted in the land, the seasons, and community rituals. Like wine itself, these themes transcend borders and bring people together, mirroring the Czecho-Italian nature of the co-authoring team.

